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Inventor(s) Okihara; Hitoshi

Syringe cap, syringe assembly, and prefilled syringe

Abstract

A syringe assembly of a prefilled syringe includes a syringe cap. A tapered section of a second tubular section constituting this cap has, on an outer peripheral surface of the tapered section, a plurality of stepped sections continuously arranged along a center axis of a cover member, so as to gradually increase a protruding amount of the tapered section from an outer peripheral surface of an annular peripheral wall section toward a vicinity of an outer edge of an opening of a first tubular section.

Inventors: **Okihara; Hitoshi (Shizuoka, JP)**

Applicant: **TERUMO KABUSHIKI KAISHA** (Tokyo, JP)

Family ID: **1000008779578**

Assignee: **TERUMO KABUSHIKI KAISHA (Tokyo, JP)**

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Primary Examiner: Tsai; Michael J

Assistant Examiner: Farrell; Kathleen Paige

Attorney, Agent or Firm: Buchanan Ingersoll & Rooney PC

Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/JP2019/012195 filed on Mar. 22, 2019, the entire content of which is incorporated herein by reference.

FIELD OF THE DISCLOSURE
(1) The present disclosure generally relates to a syringe cap, a syringe assembly, and a prefilled syringe, in which the syringe cap is detachable from a syringe body that includes a body section capable of internally accommodating a drug, and a nozzle section protruding in a distal end direction from a distal end section of the body section and having a drug discharge port at a distal

end.

BACKGROUND DISCUSSION

(2) WO 2018/061948 A discloses a syringe cap including a cap body and a tubular cover member covering the cap body. The cap body has a mounting section that is mountable to a nozzle section of a syringe body, and a viewing section located on a distal end side from the mounting section. The cover member includes a substantially opaque first tubular section and a substantially transparent second tubular section extending in the distal end direction from an opening at a distal end of the first tubular section.

(3) The cap body is at a first position where the viewing section is located in the first tube section in an unopened state of the syringe cap. At this time, the viewing section is substantially invisible (i.e., not visible). Whereas, when the mounting section of the syringe cap removed from the syringe body is brought close to the nozzle section, the cap body is displaced from the first position to a second position where the viewing section protrudes in the distal end direction from the distal end of the first tubular section, by being pushed in the distal end direction by the nozzle section. That is, the viewing section becomes substantially visible, which makes it possible to discriminate between an unopened state and an opened state of the syringe cap.

(4) FIGS. **38** to **40**B of WO 2018/061948 A describe a configuration of a syringe cap in which a tapered section inclined outward in a proximal end direction is provided at a portion on the distal end side from the first tubular section on an outer peripheral surface of the second tubular section. An outer peripheral surface of the tapered section is continuously inclined at a constant angle with respect to a center line of the cover member from a distal end to a proximal end of the tapered section, which makes it possible to prevent the syringe cap from being caught by an opening edge section of an insertion tube when the syringe assembly with the syringe cap mounted to the syringe body is inserted into the insertion tube for conveyance.

(5) However, in the syringe cap as described in WO 2018/061948 A, when the tapered section is viewed from just beside (in a direction orthogonal to a center axis of the syringe cap) in an unopened state of the syringe cap, there is a possibility that the viewing section of the cap body at the first position may be visible since light is refracted in the proximal end direction of the syringe cap on the outer peripheral surface of the tapered section.

SUMMARY

(6) A syringe cap, a syringe assembly, and a prefilled syringe are disclosed in which the syringe cap can be rather easily inserted into an insertion tube for conveyance, and an unopened state and an opened state of the syringe cap can be rather easily and reliably discriminated.

(7) In accordance with one aspect of the present disclosure, a syringe cap detachable from a syringe body is disclosed, the syringe body including: a body section capable of internally accommodating a drug; and a nozzle section protruding in a distal end direction from a distal end section of the body section and having a drug discharge port at a distal end. The syringe cap includes a cap body and a cover member having a tubular shape and covering the cap body. The cap body has a mounting section including a sealing section that liquid-tightly seals the drug discharge port, and being mountable to the nozzle section; and a viewing section located on a distal end side from the mounting section. The cover member includes: an engagement section provided on an inner peripheral surface of the cover member, and configured to engage with the cap body to prevent detachment of the cap body in a proximal end direction from the cover member; a first tubular section being substantially opaque, and having an opening at a distal end and a first space capable of internally housing the viewing section; and a second tubular section extending in a distal end direction from the opening of the first tubular section. The second tubular section includes: an annular peripheral wall section extending in a distal end direction from the opening of the first tubular section and having an outer diameter smaller than an outer diameter of the first tubular section; a distal end wall provided at a distal end of the annular peripheral wall section; a second space defined by the annular peripheral wall section, the distal end wall, and the opening of the first

tubular section; and a tapered section protruding from an outer peripheral surface of the annular peripheral wall section toward a vicinity of an outer edge of the opening of the first tubular section so as to be inclined outward in a proximal end direction. At least the tapered section and the annular peripheral wall section of the second tubular section are substantially transparent. The cap body can displace, along a center axis of the cover member in the cover member, from a first position where the viewing section is located in the first space of the first tubular section to a second position where the viewing section protrudes in a distal end direction from the distal end of the first tubular section to be arranged in the second space. An outer peripheral section of the viewing section is substantially invisible when the cap body is at the first position, and the outer peripheral section of the viewing section becomes visible when the cap body is at the second position. The sealing section is capable of liquid-tightly sealing the nozzle section in a state where the cap body is located at the first position. In a state where the cap body is arranged at the first position, when the mounting section of the syringe cap removed from the syringe body is brought close to the nozzle section of the syringe body, the mounting section is pushed in a distal end direction by the nozzle section of the syringe body so as to displace the cap body from the first position to the second position. The tapered section has, on an outer peripheral surface of the tapered section, a plurality of stepped sections continuously arranged along the center axis of the cover member, so as to gradually increase a protruding amount of the tapered section from an outer peripheral surface of the annular peripheral wall section toward a vicinity of the outer edge of the opening of the first tubular section.

(8) In accordance with another aspect of the present disclosure, a syringe assembly is disclosed, which includes: the syringe cap described above; and a syringe outer tube constituting the syringe body and being capable of accommodating a drug.

(9) In accordance with a further aspect of the present disclosure, a prefilled syringe is disclosed, which includes: the syringe assembly described above; a drug filled in the syringe outer tube; and a gasket liquid-tightly slidable in the syringe outer tube in an axial direction.

(10) According to the present disclosure, in an unopened state of the syringe cap, the cap body is at the first position where the viewing section is located in the first space of the substantially opaque first tubular section. Further, the tapered section of the second tubular section has a plurality of stepped sections on the outer peripheral surface of the tapered section of the second tubular section. As a result, when the tapered section is viewed from just beside, a refractive index of light on the outer peripheral surface of the tapered section can be reduced as compared with a case where the plurality of stepped sections are not provided on the outer peripheral surface of the tapered section. Therefore, the viewing section of the cap body at the first position can be made less visible when a user views the tapered section from just beside in an unopened state of the syringe cap.

(11) Whereas, when the mounting section of the syringe cap removed from the syringe body is brought close to the nozzle section, the cap body is displaced from the first position to the second position where the viewing section is located in the second space of the substantially transparent second tubular section. In this state, the user can visually recognize the viewing section. Therefore, even in a case where the syringe cap once removed from the syringe body is remounted to the syringe body, the user can discriminate between the unopened state and the opened state of the syringe cap.

(12) Further, the plurality of stepped sections are continuously arranged along a center axis of the cover member such that a protruding amount of the tapered section gradually increases from the outer peripheral surface of the annular peripheral wall section toward a vicinity of the outer edge of the opening of the first tubular section, which can help prevent the tapered section from being caught by the opening edge section of the insertion tube, when the syringe assembly (prefilled syringe) with the syringe cap mounted to the syringe body is inserted into the insertion tube for conveyance. Therefore, the syringe assembly (prefilled syringe) can be rather easily inserted into the insertion tube for conveyance.

(13) In accordance with an aspect, a syringe cap is disclosed configured to be detachable from and attachable to a syringe body, the syringe cap comprising: a cap body, the cap body includes: a mounting section including a sealing section configured to seal a drug discharge port on the syringe body and configured to be mounted to a nozzle section of the syringe body; and a viewing section located on a distal end side from the mounting section; a cover member having a tubular shape and covering the cap body, the cover member includes: an engagement section provided on an inner peripheral surface of the cover member, and the cover member being configured to engage with the cap body; a first tubular section having an opening at a distal end and a first space capable of internally housing the viewing section; and a second tubular section extending in a distal end direction from the opening of the first tubular section; the second tubular section includes: an annular peripheral wall section extending in a distal end direction from the opening of the first tubular section and having an outer diameter smaller than an outer diameter of the first tubular section; a distal end wall provided at a distal end of the annular peripheral wall section; a second space defined by the annular peripheral wall section, the distal end wall, and the opening of the first tubular section; and a tapered section protruding from an outer peripheral surface of the annular peripheral wall section toward a vicinity of an outer edge of the opening of the first tubular section so as to be inclined outward in a proximal end direction; and the tapered section has, on an outer peripheral surface of the tapered section, a plurality of stepped sections continuously arranged along the center axis of the cover member, so as to gradually increase a protruding amount of the tapered section from an outer peripheral surface of the annular peripheral wall section toward a vicinity of the outer edge of the opening of the first tubular section.
(14) In accordance with another aspect, a syringe assembly is disclosed, the syringe assembly comprising: syringe body, the syringe body including a body section configured to accommodate a drug, and a nozzle section protruding in a distal end direction from a distal end section of the body section and having a drug discharge port at a distal end; and a syringe cap configured to be detachable from and attachable to the syringe body, the syringe cap comprising: a cap body, the cap body includes: a mounting section including a sealing section configured to seal a drug discharge port on the syringe body and configured to be mounted to a nozzle section of the syringe body; and a viewing section located on a distal end side from the mounting section; a cover member having a tubular shape and covering the cap body, the cover member includes: an engagement section provided on an inner peripheral surface of the cover member, and the cover member being configured to engage with the cap body; a first tubular section having an opening at a distal end and a first space capable of internally housing the viewing section; and a second tubular section extending in a distal end direction from the opening of the first tubular section; the second tubular section includes: an annular peripheral wall section extending in a distal end direction from the opening of the first tubular section and having an outer diameter smaller than an outer diameter of the first tubular section; a distal end wall provided at a distal end of the annular peripheral wall section; a second space defined by the annular peripheral wall section, the distal end wall, and the opening of the first tubular section; and a tapered section protruding from an outer peripheral surface of the annular peripheral wall section toward a vicinity of an outer edge of the opening of the first tubular section so as to be inclined outward in a proximal end direction; and the tapered section has, on an outer peripheral surface of the tapered section, a plurality of stepped sections continuously arranged along the center axis of the cover member, so as to gradually increase a protruding amount of the tapered section from an outer peripheral surface of the annular peripheral wall section toward a vicinity of the outer edge of the opening of the first tubular section.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a longitudinal cross-sectional view of a prefilled syringe according to an embodiment of the present disclosure.

(2) FIG. **2** is a partially omitted enlarged cross-sectional view of a distal end side of the prefilled syringe of FIG. **1**.

(3) FIG. **3** is an exploded perspective view of FIG. **2**.

(4) FIG. **4** is an enlarged view of a tapered section.

(5) FIG. **5**A is a first explanatory view of a recapping operation of a syringe cap, and FIG. **5**B is a second explanatory view of the recapping operation of the syringe cap.

(6) FIG. **6** is an explanatory cross-sectional view illustrating a state where the prefilled syringe is inserted into an insertion tube for conveyance.

(7) FIG. **7** is a partially omitted enlarged view of a tapered section according to a modification.

DETAILED DESCRIPTION

(8) Set forth below with reference to the accompanying drawings is a detailed description of embodiments of a syringe cap, a syringe assembly, and a prefilled syringe, in which the syringe cap is detachable from a syringe body that includes a body section capable of internally accommodating a drug, and a nozzle section protruding in a distal end direction from a distal end section of the body section and having a drug discharge port at a distal end. Note that since embodiments described below are preferred specific examples of the present disclosure, although various technically preferable limitations are given, the scope of the present disclosure is not limited to the embodiments unless otherwise specified in the following descriptions. In the following description of the prefilled syringe and components of the prefilled syringe, the left side in FIG. **1** is referred to as a “distal end”, and the right side is referred to as a “proximal end”.

(9) As illustrated in FIG. **1**, a prefilled syringe **12** includes a syringe body **14** and a syringe cap **10** (hereinafter, may be simply referred to as a “cap **10**”) detachably attachable to the syringe body **14**. The syringe body **14** includes a syringe outer tube **16**, a gasket **18** slidably inserted into the syringe outer tube **16**, and a plunger **20** connected to the gasket **18**. In the present embodiment, the syringe outer tube **16** and the cap **10** constitute a syringe assembly **22**, and the prefilled syringe **12** is assembled by inserting the gasket **18** connected with the plunger **20** in a state where the syringe outer tube **16** of the syringe assembly **22** is filled with a drug M.

(10) The syringe outer tube **16** can include: a cylindrical body section **24** extending in an axial direction; a hollow nozzle section **26** projecting in the distal end direction from a distal end section of the body section **24**; a syringe-side connecting section **28** provided on an outer peripheral side of the nozzle section **26**; and a flange section **30** provided at a proximal end section of the body section **24**. The body section **24**, the nozzle section **26**, the syringe-side connecting section **28**, and the flange section **30** are integrally formed (i.e., connected together so as to make up a single complete piece or unit).

(11) As illustrated in FIG. **2**, the nozzle section **26** is a cylindrical member provided with a drug discharge port **26***a* at a distal end, and is configured as a luer connector. The syringe-side connecting section **28** is a lock adapter that projects from a distal end section of the body section **24** to a distal end side concentrically with the nozzle section **26**, and has a female screw section **32** formed on an inner peripheral surface. To the syringe-side connecting section **28**, the cap **10**, an injection needle, and the like are detachably attachable.

(12) A constituent material of the syringe outer tube **16** is not particularly limited, but is preferably form by, for example, polyolefin such as polypropylene, polyurethane, polyethylene, cyclic polyolefin, or polymethylpentene **1**, a resin material such as polyester, nylon, polycarbonate, polymethyl methacrylate (PMMA), polyetherimide (PEI), polyethersulfone, polyether ether ketone (PEEK), fluororesin, polyphenylene sulfide (PPS), or a polyacetal resin (POM), a metal material such as stainless steel, glass, or the like.

(13) In FIG. **1**, the gasket **18** is liquid-tightly slidable in the syringe outer tube **16** in an axial direction, and feeds the drug M filled in the syringe outer tube **16**. A distal end of the plunger **20** is connected to the gasket **18**.
(14) As illustrated in FIGS. **2** and **3**, the cap **10** includes a cap body **34** that closes the drug discharge port **26***a* of the nozzle section **26** in an unopened state, and a tubular cover member **36** that covers the cap body **34**.
(15) Examples of a constituent material of the cap body **34** can include rubber, a synthetic resin elastomer, and the like. Examples of the rubber can include isoprene rubber, butyl rubber, latex rubber, and silicone rubber. As the synthetic resin elastomer, for example, a styrene elastomer, an olefin elastomer, or the like can be used.
(16) The cap body **34** includes a mounting tube section **38** (mounting section) and a viewing section **40** (distal end extending section) extending the distal end direction from a distal end of the mounting tube section **38**. The mounting tube section **38** includes a sealing section **42** that is located at a distal end section of the mounting tube section **38** and is capable of liquid-tightly sealing the drug discharge port **26***a*, and a raised section **44** raised from the sealing section **42**. The sealing section **42** seals the drug discharge port **26***a* by abutting on a distal end section of the nozzle section **26**, specifically, a distal end surface of the nozzle section **26** or a side peripheral surface of the distal end section of the nozzle section **26**.
(17) The liquid-tightly sealing the drug discharge port **26***a* means that the drug M does not leak to the outside of the cap body **34**. Therefore, even when the sealing section **42** abuts only on the side peripheral surface of the distal end section of the nozzle section **26**, the sealing section **42** liquid-tightly seals the drug discharge port **26***a*.
(18) At the proximal end section of an inner peripheral surface of the mounting tube section **38**, an annular abutting protruding section **46** (abutting section) capable of abutting on a distal end section of the nozzle section **26** is provided. In a mounted state (unopened state) in which the cap body **34** is mounted to the nozzle section **26**, the abutting protruding section **46** is in contact with an outer peripheral surface of the nozzle section **26** in a compressed and deformed state.
(19) On an outer peripheral surface of the mounting tube section **38**, an annular bulging section **48** is provided. The bulging section **48** is a section having a maximum outer diameter of the cap body **34**. The bulging section **48** can be compressed and deformed by being sandwiched between the nozzle section **26** and a tubular connecting section **54**, in the mounted state of the cap body **34**.
(20) The viewing section **40** is formed in a cylindrical shape. The viewing section **40** may be colored in a color that is relatively easily recognized by the user, such as red, for example. The coloring of the viewing section **40** may be performed by applying paint to an outer surface of the viewing section **40**, or the cap body **34** may be made of rubber or synthetic resin colored in advance.
(21) The cover member **36** includes a cap cover **50** and a distal end cover member **52**.
(22) The cap cover **50** is formed in a cylindrical shape and can be made of a resin material having no transparency (substantially opaque resin material). The cap cover **50** can include: the tubular connecting section **54** that is located at a proximal end section of cap cover **50** and is detachably attachable to a syringe side by screwing; a tubular section **56** extending in the distal end direction from a distal end of the tubular connecting section **54**; and an opening **58** provided at a distal end of the tubular section **56** to expose the cap body **34** from the cap cover **50**.
(23) At a proximal end section of an inner peripheral surface of the tubular connecting section **54**, a plurality of engagement claw sections **60** (engagement sections, engagement protrusions) extending radially inward are provided. The plurality of engagement claw sections **60** are intermittently provided along a circumferential direction of the tubular connecting section **54**. Specifically, the plurality of engagement claw sections **60** are provided at equal intervals along the circumferential direction of the tubular connecting section **54**. An inner diameter of a hole formed by projecting ends (inner end sections) of the plurality of engagement claw sections **60** is smaller than an outer

diameter of a proximal end section of the mounting tube section **38** of the cap body **34**.
(24) In accordance with an exemplary embodiment, in each of the engagement claw sections **60**, a surface directed in the distal end direction is a flat surface extending in a direction orthogonal to a center axis Ax of the cover member **36**, and can be in contact with a proximal end surface of the cap body **34**. Furthermore, the engagement claw section **60** has an inclined surface inclined inward in the distal end direction on the proximal end side of the engagement claw section **60**. As a result, when the cap body **34** is inserted into the cap cover **50** from the proximal end of the cap cover **50**, the cap body **34** can relatively easily get over the engagement claw section **60**.
(25) A proximal end of the tubular section **56** is provided with an insertion regulating section **62** that can abut on a distal end of the syringe-side connecting section **28**. The insertion regulating section **62** helps regulate an insertion length of the tubular connecting section **54** between the syringe-side connecting section **28** and the nozzle section **26**, by abutting on the distal end of the syringe-side connecting section **28**. The tubular connecting section **54** is a cylindrical member provided concentrically with the cap cover **50**, and has an outer peripheral surface provided with a male screw section **64** that can be screwed with the female screw section **32**.
(26) The tubular section **56** is formed in a size that allows a user to rather easily pick up the tubular section **56** with the fingers of the user. At a proximal end section of an outer peripheral surface of the tubular section **56**, an annular recess **66** is formed. On a bottom surface of the annular recess **66**, two through holes **68** are formed. In accordance with an exemplary embodiment, the two through holes **68** are opposed to each other.
(27) On the distal end side from the annular recess **66** on the outer peripheral surface of the tubular section **56**, an anti-slip section **70** functioning as a slip stopper for the user's fingers is formed. The anti-slip section **70** can be formed, for example, by providing a plurality of ribs **72** extending in an axial direction at equal intervals in a circumferential direction. In the present embodiment, for example, six (6) ribs **72** are provided in the circumferential direction of the tubular section **56**. As described above, by setting the number of ribs **72** to six, moldability (injection molding accuracy) of the cap cover **50** can be improved.
(28) On the distal end side of the annular recess **66** on an inner peripheral surface of the tubular section **56**, a locking protruding section **74** to lock the distal end cover member **52** protrudes radially inward. Two locking protruding sections **74** are provided so as to be opposed to each other.
(29) The distal end cover member **52** is formed to have a substantially U-shaped longitudinal cross section, and is to cover the cap body **34** such that the user operating the cap **10** together with the cap cover **50** cannot touch the cap body **34**. That is, the distal end cover member **52** has a contact prevention function. The distal end cover member **52** also functions as a detachment prevention section that help prevent detachment of the cap body **34** from the opening **58** of the cap cover **50**.
(30) The distal end cover member **52** can include: an annular section **76** having a proximal end section fitted to the tubular section **56** of the cap cover **50** so as to project toward a distal end side from the opening **58** of the cap cover **50**; and a distal end wall **78** provided at a distal end section of the annular section **76**.
(31) The annular section **76** can include an annular peripheral wall section **80** on the distal end side and a tubular extending inner tube section **82** extending in the proximal end direction from the annular peripheral wall section **80**. In accordance with an exemplary embodiment, an inner diameter of the annular section **76** is constant from the distal end to the proximal end, and is larger than an outer diameter of the viewing section **40**. In other words, an inner diameter of the distal end cover member **52** is substantially uniform from a distal end of the annular peripheral wall section **80** to a proximal end of the extending inner tube section **82**.
(32) The proximal end of the extending inner tube section **82** is in contact with the distal end of the tubular connecting section **54**. At a portion corresponding to the annular recess **66** of the cap cover **50** on an outer peripheral surface of the extending inner tube section **82**, an annular locking claw **84** to be in contact with the locking protruding section **74** of the cap cover **50** is provided. In

accordance with an exemplary embodiment, an outer diameter of the locking claw **84** is larger than a separation interval of the locking protruding section **74**.

(33) On an outer peripheral surface of the annular peripheral wall section **80**, there is provided a tapered section **86** protruding so as to be inclined outward in the proximal end direction toward the distal end of the tubular section **56** (in a vicinity of an outer edge of the opening **58** of the tubular section **56**). The tapered section **86** annularly extends along a circumferential direction of the annular peripheral wall section **80**.

(34) As shown in FIGS. **2** to **4**, on an outer peripheral surface of the tapered section **86**, a plurality of (for example, seven) stepped sections **88** are arranged continuously along the center axis Ax of the cover member **36** such that a protruding amount of the tapered section **86** gradually increases from the outer peripheral surface of the annular peripheral wall section **80** toward the distal end of the tubular section **56**. Each stepped section **88** annularly extends along the circumferential direction of the annular peripheral wall section **80**.

(35) In accordance with an exemplary embodiment, each of the stepped sections **88** includes: a first surface **90**; a bent section **92** located at an outer edge of the first surface **90**; a second surface **94** extending in the proximal end direction from the bent section **92**; and an inclined surface **96** extending while being inclined outward (radially outward) from a proximal end of the second surface **94** and continuing to an inner edge of the first surface **90** of the stepped section **88** adjacent on the proximal end side.

(36) In accordance with an exemplary embodiment, an angle θ**1** formed by the first surface **90** and a parallel straight line L**1** parallel to the center axis Ax of the cover member **36** is larger than a taper angle θ**2** formed by the parallel straight line L**1** and a virtual straight line L**2** passing through a distal end and a proximal end of the tapered section **86**. The second surface **94** extends parallel to the center axis Ax of the cover member **36**.

(37) The angle θ**1** is preferably set, for example, to 60° or more and 90° or less (i.e., 60° to 90°). In the present embodiment, for example, the angle θ**1** is set to 90°. The taper angle θ**2** can be preferably be set, for example, to 35° or more and 55° or less (i.e., 35° to 55°).

(38) In accordance with an exemplary embodiment, the inclined surface **96** extends along the virtual straight line L**2**. In other words, the inclined surface **96** extends so as to overlap with the virtual straight line L**2**. Note that the inclined surface **96** may extend parallel to the virtual straight line L**2**. The bent section **92** has a curved surface protruding outward.

(39) In accordance with an exemplary embodiment, a size, a number, a position, and a shape of the stepped section **88** can be appropriately set. For example, the stepped section **88** does not need to have the inclined surface **96**. That is, the stepped section **88** may be formed such that a proximal end of the second surface **94** is continuous to an inner edge of the first surface **90** of the stepped section **88** adjacent on the proximal end side of the adjacent stepped section **88**.

(40) In FIGS. **2** and **3**, between an outer edge of the distal end wall **78** and the annular peripheral wall section **80** (boundary section), a distal end curved surface **98** protruding outward is formed. A curvature radius of the distal end curved surface **98** can be preferably set, for example, to 0.3 mm or more and 1.0 mm or less (i.e., 0.3 mm to 1.0 mm).

(41) In accordance with an exemplary embodiment, the distal end cover member **52** can be integrally molded using a resin material having transparency. In a case where the cap cover **50** is made of a resin material having transparency, the transparency of the distal end cover member **52** is set higher than the transparency of the cap cover **50**, which allows the user to visually recognize the inside of the distal end cover member **52** from the outside of the cap **10** more clearly than the inside of the cap cover **50**. In the present embodiment, the distal end cover member **52** can be colorless, but may be colored.

(42) As illustrated in FIG. **2**, such a cover member **36** includes the engagement claw section **60**, a first tubular section **100**, and a second tubular section **102**. The engagement claw section **60** is provided on an inner peripheral surface of the cover member **36**, and helps prevent detachment of

the cap body **34** in the proximal end direction from the cover member **36**, by engaging with the cap body **34**. The first tubular section **100** has the opening **58** at a distal end and a first space S**1** that can internally accommodate the viewing section **40**. In accordance with an exemplary embodiment, the first tubular section **100** is substantially opaque. The first tubular section **100** includes the extending inner tube section **82** forming an inner peripheral section of the first tubular section **100**, and the tubular section **56** forming an outer peripheral section of the first tubular section **100**.

(43) The second tubular section **102** is substantially transparent as a whole, and extends in the distal end direction from the opening **58** of the first tubular section **100**. The second tubular section **102** includes the annular peripheral wall section **80**, the distal end wall **78**, a second space S**2**, and the tapered section **86**. The annular peripheral wall section **80** extends in the distal end direction from the opening **58** of the first tubular section **100**, and has an outer diameter smaller than an outer diameter of the first tubular section **100**. The distal end wall **78** is provided at the distal end of the annular peripheral wall section **80**. The second space S**2** is defined by the annular peripheral wall section **80**, the distal end wall **78**, and the opening **58** of the first tubular section **100**. The tapered section **86** protrudes so as to be inclined outward in the proximal end direction from the outer peripheral surface of the annular peripheral wall section **80** toward a vicinity of the outer edge of the opening **58** of the first tubular section **100**.

(44) In a prefilled syringe **12** as disclosed, in an unopened state of the cap **10**, the cap body **34** is at a first position where the viewing section **40** is located in the first space S**1** of the first tubular section **100**. Specifically, the distal end of the cap body **34** is located in the proximal end direction from the opening **58** of the tubular section **56**, and inside the extending inner tube section **82** of the distal end cover member **52**. As a result, the outer peripheral section of the viewing section **40** can be hidden by the substantially opaque first tubular section (tubular section **56**) and cannot be visually recognized from the outside. Further, the mounting tube section **38** is in a state of being mounted to the nozzle section **26**, and the sealing section **42** of the mounting tube section **38** liquid-tightly seals the drug discharge port **26***a* of the nozzle section **26**.

(45) In a case of opening the cap **10** from the syringe body **14**, the cover member **36** is pulled out from the syringe outer tube **16** in a state where screwing between the male screw section **64** and the female screw section **32** is released, which causes the engagement claw section **60** of the tubular connecting section **54** to come into contact with the proximal end surface of the cap body **34**. Then, the cap body **34** is pushed in the distal end direction by the tubular connecting section **54**, and the mounting tube section **38** is detached from the nozzle section **26**, which leads to opening of the cap **10** (see FIG. 5A).

(46) In a case of recapping the opened cap **10** to the syringe outer tube **16**, the distal end section of the nozzle section **26** of the syringe outer tube **16** is inserted into the tubular connecting section **54** from the opening **58** on the proximal end side of the tubular connecting section **54**, which causes the distal end section of the nozzle section **26** to come into contact with the abutting protruding section **46** of the cap body **34**.

(47) Subsequently, when the cap cover **50** and the syringe outer tube **16** are brought relatively close to each other, by the abutting protruding section **46** of the cap body **34** abutting with the distal end section of the nozzle section **26**, the cap body **34** at the first position is displaced in the distal end direction with respect to the cap cover **50** by being pushed by the nozzle section **26**, and the viewing section **40** protrudes toward the distal end side from the opening **58** on the distal end side of the cap cover **50**.

(48) Then, by screwing the male screw section **64** of the tubular connecting section **54** with the female screw section **32** of the syringe-side connecting section **28**, the opened cap **10** is recapped (remounted) to the syringe outer tube **16**. At this time, the cap body **34** is at the second position where the viewing section **40** is located in the second space S**2** of the second tubular section **102** (on the distal end side from the opening **58** of the tubular section **56**) (see FIG. 5B). Therefore, the outer peripheral section of the viewing section **40** can be visually recognized from the outside via

the transparent second tubular section **102** (distal end cover member **52**).
(49) In this case, the cap **10**, the syringe assembly **22**, and the prefilled syringe **12** according to the present embodiment have the following effects.
(50) In the cap **10**, in an unopened state of the cap **10**, the cap body **34** is at the first position where the viewing section **40** is located in the first space S**1** of the substantially opaque first tubular section **100**. Further, the tapered section **86** of the second tubular section **102** has a plurality of stepped sections **88** on the outer peripheral surface of the tapered section **86** of the second tubular section **102**. As a result, when the tapered section **86** is viewed from just beside (a direction orthogonal to the center axis Ax of the cover member **36**), a refractive index of light on the outer peripheral surface of the tapered section **86** toward the proximal end side of the cover member **36** can be reduced as compared with a case where the plurality of stepped sections **88** are not provided on the outer peripheral surface of the tapered section **86**. Therefore, the viewing section **40** of the cap body **34** at the first position can be made less visible when the user views the tapered section **86** from just beside in an unopened state of the cap **10**.
(51) Whereas, when the mounting tube section **38** of the cap **10** removed from the syringe body **14** is brought close to the nozzle section **26**, the cap body **34** is displaced from the first position to the second position where the viewing section **40** is located in the second space S**2** of the substantially transparent second tubular section **102**. In this state, the user can visually recognize the viewing section **40**. Therefore, even in a case where the cap **10** once removed from the syringe body **14** is remounted to the syringe body **14**, the user can discriminate between the unopened state and the opened state of the cap **10**.
(52) Further, the plurality of stepped sections **88** can be arranged continuously along the center axis Ax of the cover member **36** such that a protruding amount of the tapered section **86** gradually increases from the outer peripheral surface of the annular peripheral wall section **80** toward a vicinity of the outer edge of the opening **58** of the first tubular section **100**. As a result, as shown in FIG. **6**, it is possible to prevent the tapered section **86** from being caught by an opening edge section **202** of an insertion tube **200** when the syringe assembly **22** (prefilled syringe **12**) is inserted into the insertion tube **200** for conveyance. Therefore, the syringe assembly **22** can be rather easily inserted into the insertion tube **200** for conveyance.
(53) In the cap **10**, each of the plurality of stepped sections **88** has the first surface **90**, the bent section **92** located at the outer edge of the first surface **90**, and the second surface **94** extending in the proximal end direction from the bent section **92**. The angle θ**1** formed by the first surface **90** and the parallel straight line L**1** parallel to the center axis Ax of the cover member **36** is larger than the taper angle θ**2** formed by the parallel straight line L**1** and the virtual straight line L**2** passing through the distal end and the proximal end of the tapered section **86**. The second surface **94** extends parallel to the center axis Ax.
(54) According to such a configuration, when the user views the tapered section **86** from just beside, a refractive index of light on the second surface **94** of the tapered section **86** can be effectively reduced. Therefore, the viewing section **40** of the cap body **34** at the first position can be made further less visible when the user views the tapered section **86** from just beside in an unopened state of the cap **10**.
(55) In accordance with an exemplary embodiment, the taper angle θ**2** can be, for example, 35° or more and 55° or less (i.e., 35° to 55°).
(56) According to such a configuration, the syringe assembly **22** (prefilled syringe **12**) can be rather easily inserted into the insertion tube **200** for conveyance.
(57) Each of the plurality of stepped sections **88** includes the inclined surface **96** extending while being inclined outward from the proximal end of the second surface **94** and continuing to the inner edge of the first surface **90** of the stepped section **88** adjacent on the proximal end side.
(58) According to such a configuration, the tapered section **86** can be effectively prevented from being caught by the opening edge section **202** of the insertion tube **200**.

(59) The bent section **92** has a curved surface protruding outward.
(60) According to such a configuration, the tapered section **86** can be effectively prevented from being caught by the opening edge section **202** of the insertion tube **200**.
(61) In accordance with an exemplary embodiment, the second tubular section **102** is substantially transparent as a whole, and has the distal end curved surface **98** projecting outward between the outer edge of the distal end wall **78** and the annular peripheral wall section **80**. A curvature radius of the distal end curved surface **98** is 0.3 mm or more and 1.0 mm or less (i.e., 0.3 mm to 1.0 mm).
(62) According to such a configuration, since the entire second tubular section **102** is transparent, the user can rather easily visually recognize the viewing section **40** in a state where the cap **10** is remounted to the syringe body **14**. In addition, it is possible to suppress a decrease in visibility of the viewing section **40** due to reflection of light on the distal end curved surface **98**.
(63) The cover member **36** includes the tubular section **56** constituting at least an outer peripheral section of the first tubular section **100** and being substantially opaque, and the cover member **36** having the second tubular section **102** and being substantially transparent.
(64) According to such a configuration, the configuration of the substantially opaque first tubular section **100** and the substantially transparent second tubular section **102** can be simplified.
(65) The distal end cover member **52** includes the extending inner tube section **82** that extends from the proximal end of the annular peripheral wall section **80** into the tubular section **56** and forms the inner peripheral section of the first tubular section **100**. An inner diameter of the distal end cover member **52** is substantially uniform from the distal end of the annular peripheral wall section **80** to a proximal end of the extending inner tube section **82**.
(66) According to such a configuration, since a step is not substantially formed on the inner peripheral surface of the cover member **36** from the first space S**1** to the second space S**2**, the cap body **34** can be rather smoothly displaced from the first position to the second position.
(67) The cap **10**, the syringe assembly **22**, and the prefilled syringe **12** according to the present disclosure are not limited to the above-described embodiment, and it is a matter of course that various configurations can be adopted without departing from the gist of the present disclosure.
(68) As illustrated in FIG. **7**, a second surface **110** of the tapered section **86** may be inclined outward (radially outward) from the bent section **92** in the proximal end direction. In this case, an angle θ**1** (first angle θ**1**) formed by the first surface **90** and the parallel straight line L**1** parallel to the center axis Ax of the cover member **36** is larger than a taper angle θ**2**. An angle θ**3** (second angle θ**3**) formed by the second surface **110** and a parallel straight line L**3** parallel to the center axis Ax of the cover member **36** is smaller than the taper angle θ**2**.
(69) That is, the first angle θ**1**, the taper angle θ**2**, and the second angle θ**3** are set such that a relationship of θ**3**<Θ**2**<Θ**1** is established. The first angle θ**1** is preferably set, for example, to 60° or more and 90° or less (i.e., 60° to 90°). The taper angle θ**2** is preferably set, for example, to 35° or more and 55° or less (i.e., 35° to 55°). The second angle θ**3** is preferably set, for example, to 30° or less.
(70) In the present embodiment, each of the plurality of stepped sections **88** has the first surface **90**, the bent section **92** located at the outer edge of the first surface **90**, and the second surface **110** extending in the proximal end direction from the bent section **92**. The first angle θ**1** is larger than the taper angle θ**2**, and the second angle θ**3** is smaller than the taper angle θ**2**.
(71) According to such a configuration, when the user views the tapered section **86** from just beside, a refractive index of light on the second surface **110** of the tapered section **86** can be effectively reduced. Therefore, the viewing section **40** of the cap body **34** at the first position can be made further less visible when the user views the tapered section **86** from just beside in an unopened state of the cap **10**.
(72) The present disclosure may be, for example, a prefilled syringe **12** having no plunger **20** (i.e., without a plunger **20**). In this case, to the prefilled syringe **12**, a pressing member that presses the gasket **18** in the distal end direction is separately mounted. In addition, the cap **10**, the syringe

assembly **22**, or the prefilled syringe **12** in which the abutting protruding section **46** is omitted from the cap body **34** may be adopted. In this case, by making an inner diameter of the opening **58** on the proximal end side of the mounting tube section **38** smaller than a distal end outer diameter of the nozzle section **26** of the syringe outer tube **16**, the proximal end of the mounting tube section **38** functions as an abutting section that can abut on the distal end section of the nozzle section **26**.
(73) Further, the cap body **34** may be configured such that the viewing section **40** does not include the distal end of the cap body **34**, and the distal end of the cap body **34** protrudes to the distal end side from the opening **58** of the cap cover **50**, at the first position. In this case, the viewing section **40** of the cap body **34** is provided so as to be located in the tubular section **56** of the cap cover **50** when the cap body **34** is at the first position, and to protrude from the opening **58** of the cap cover **50** when the cap body **34** is at the second position.
(74) Examples of a viewing section **40** can include: a viewing section **40** formed by coloring an outer peripheral surface that is slightly on the proximal end side from the distal end of the cap body **34** in red or the like; and a viewing section **40** formed by a colored member fitted so as to cover an outer peripheral section that is slightly on the proximal end side from the distal end of the distal end protruding section of the cap body **34**. Even with such a configuration, the user can rather easily and reliably discriminate between the unopened state and the recapped state of the cap **10**.
(75) In the distal end wall **78** of the distal end cover member **52**, a hole having such a size that a user's finger cannot be inserted may be formed. In a case where such a hole is formed on the distal end wall **78**, by inserting a jig into the hole of the distal end wall **78** and pushing the cap body **34** into the nozzle section **26**, the previously assembled cap **10** can be mounted to the syringe outer tube **16**.
(76) Further, the cap **10** may be provided with a temporary fixing mechanism that performs temporary fixing to help prevent movement of the cap body **34** from the first position to the second position in a state where the cap **10** is removed from the syringe body **14**. Examples of such a temporary fixing mechanism can include a temporary fixing protrusion provided on an inner peripheral surface of the cap cover **50** and engaged with a part of the cap body **34** arranged at the first position. In this case, an engagement force between the temporary fixing protrusion and the cap body **34** is set to be releasable by pressing the mounting section (mounting tube section **38**) of the cap body **34** with the nozzle section **26** of the syringe body **14**, which can help prevent the cap body **34** from being unintentionally displaced from the first position to the second position in the unopened state, and can displace the cap body **34** from the first position to the second position when the mounting section (mounting tube section **38**) of the cap **10** removed from the syringe body **14** is brought close to the nozzle section **26** of the syringe body **14** in a state where the cap body **34** is arranged at the first position.
(77) In accordance with an exemplary embodiment, in a state where the cap **10** is removed from the syringe body **14**, the cap body **34** is movable from the first position to the second position without the mounting section (mounting tube section **38**) being pushed by the nozzle section **26**. Therefore, in a case where the cap **10** removed from the syringe body **14** is remounted to the syringe body **14** in a state where the cap body **34** is arranged at the second position, displacement of the cap body **34** from the first position to the second position does not occur. Even in this case, when the mounting section (mounting tube section **38**) of the cap body **34** arranged at the second position abuts on the nozzle section **26**, displacement of the cap body **34** from the second position to the first position can be regulated (or controlled), and the viewing section **40** is located at the second position. Therefore, the user can rather easily and reliably discriminate between the unopened state and the opened state of the cap **10**.
(78) In the above embodiment, a syringe cap **10** is disclosed that is detachable from a syringe body **14** including: a body section **24** capable of internally accommodating a drug M; and a nozzle section **26** protruding in a distal end direction from a distal end section of the body section **24** and having a drug discharge port **26***a* at a distal end, in which the syringe cap **10** includes: a cap body

34; and a tubular cover member **36** covering the cap body **34**, the cap body **34** includes: a mounting section **38** having a sealing section **42** that liquid-tightly seals the drug discharge port **26***a*, and being mountable to the nozzle section **26**; and a viewing section **40** located on a distal end side from the mounting section **38**, the cover member **36** includes: an engagement section **60** provided on an inner peripheral surface of the cover member **36**, and configured to engage with the cap body **34** to prevent detachment of the cap body **34** in a proximal end direction from the cover member **36**; a first tubular section **100** being substantially opaque, and having an opening **58** at a distal end and a first space S**1** capable of internally housing the viewing section **40**; and a second tubular section **102** extending in a distal end direction from the opening **58** of the first tubular section **100**, the second tubular section **102** includes: an annular peripheral wall section **80** extending in a distal end direction from the opening **58** of the first tubular section **100** and having an outer diameter smaller than an outer diameter of the first tubular section **100**; a distal end wall **78** provided at a distal end of the annular peripheral wall section **80**; a second space S**2** defined by the annular peripheral wall section **80**, the distal end wall **78**, and the opening **58** of the first tubular section **100**; and a tapered section **86** protruding from an outer peripheral surface of the annular peripheral wall section **80** toward a vicinity of an outer edge of the opening **58** of the first tubular section **100** so as to be inclined outward in a proximal end direction, at least the tapered section **86** and the annular peripheral wall section **80** of the second tubular section **102** are substantially transparent, the cap body **34** can displace, along a center axis Ax of the cover member **36** in the cover member **36**, from a first position where the viewing section **40** is located in the first space S**1** of the first tubular section **100** to a second position where the viewing section **40** protrudes in a distal end direction from the distal end of the first tubular section **100** to be arranged in the second space S**2**, an outer peripheral section of the viewing section **40** is substantially invisible when the cap body **34** is at the first position, and the outer peripheral section of the viewing section **40** becomes visible when the cap body **34** is at the second position, the sealing section **42** is capable of liquid-tightly sealing the nozzle section **26** in a state where the cap body **34** is located at the first position, in a state where the cap body **34** is arranged at the first position, when the mounting section **38** of the syringe cap **10** removed from the syringe body **14** is brought close to the nozzle section **26** of the syringe body **14**, the mounting section **38** is pushed in a distal end direction by the nozzle section **26** of the syringe body **14** so as to displace the cap body **34** from the first position to the second position, and the tapered section **86** has, on an outer peripheral surface of the tapered section **86**, a plurality of stepped sections **88** continuously arranged along the center axis Ax of the cover member **36**, so as to gradually increase a protruding amount of the tapered section **86** from an outer peripheral surface of the annular peripheral wall section **80** toward a vicinity of the outer edge of the opening **58** of the first tubular section **100**.

(79) In the syringe cap **10** described above, each of the plurality of stepped sections **88** may have a first surface **90**, a bent section **92** located at an outer edge of the first surface **90**, and a second surface **94** extending in a proximal end direction from the bent section **92**. A first angle θ**1** formed by the first surface **90** and a parallel straight line L**1** parallel to the center axis Ax of the cover member **36** may be larger than a taper angle θ**2** formed by the parallel straight line L**1** and a virtual straight line L**2** passing through a distal end and a proximal end of the tapered section **86**, and a second angle θ**3** formed by the second surface **94** and a parallel straight line L**3** parallel to the center axis Ax of the cover member **36** may be smaller than the taper angle θ**2**.

(80) In the syringe cap **10** described above, each of the plurality of stepped sections **88** may have a first surface **90**, a bent section **92** located at an outer edge of the first surface **90**, and a second surface **94** extending in a proximal end direction from the bent section **92**. An angle θ**1** formed by the first surface **90** and a parallel straight line L**1** parallel to the center axis Ax of the cover member **36** may be larger than a taper angle θ**2** formed by the parallel straight line L**1** and a virtual straight line L**2** passing through the distal end and the proximal end of the tapered section **86**, and the second surface **94** may extend parallel to the center axis Ax.

(81) In the syringe cap **10** described above, the taper angle θ**2** may be, for example, 35° or more and 55° or less (i.e., 35° to 55°).
(82) In the syringe cap **10** described above, the first angle θ**1** may be, for example, 60° or more and 90° or less (i.e., 60° to 90°), and the second angle θ**3** may be, for example, 30° or less.
(83) In the syringe cap **10** described above, each of the plurality of stepped sections **88** may have an inclined surface **96** extending while being inclined outward from a proximal end of the second surface **94** and continuing to an inner edge of the first surface **90** of a stepped section **88** adjacent on the proximal end side.
(84) In the syringe cap **10** described above, the bent section **92** may have a curved surface protruding outward.
(85) In the syringe cap **10** described above, the second tubular section **102** may be substantially transparent as a whole and have a distal end curved surface **98** protruding outward between an outer edge of the distal end wall **78** and the annular peripheral wall section **80**, and a curvature radius of the distal end curved surface **98** may be, for example, 0.3 mm or more and 1.0 mm or less (i.e., 0.3 mm to 1.0 mm).
(86) In the syringe cap **10** described above, the cover member **36** may include a tubular section **56** constituting at least an outer peripheral section of the first tubular section **100** and being substantially opaque, and a distal end cover member **52** having the second tubular section **102** and being substantially transparent.
(87) In the syringe cap **10** described above, the distal end cover member **52** may have an extending inner tube section **82** that extends from a proximal end of the annular peripheral wall section **80** into the tubular section **56** and forms an inner peripheral section of the first tubular section **100**, and an inner diameter of the distal end cover member **52** may be substantially uniform from a distal end of the annular peripheral wall section **80** to a proximal end of the extending inner tube section **82**.
(88) In the syringe cap **10** described above, the syringe body **14** may include a lock adapter **28** having a tubular shape, having a female screw section **32** on an inner peripheral surface, and covering an outer peripheral section of the nozzle section **26**, the mounting section **38** may be a mounting tube section **38** having a tubular shape and being capable of accommodating the nozzle section **26**, the mounting tube section **38** may have: the sealing section **42** located at a distal end section of the mounting tube section **38**; and an abutting section **46** located at a proximal end section of the mounting tube section **38** and being capable of abutting on a distal end section of the nozzle section **26**, the cover member **36** may have a tubular connecting section **54** that extends in a proximal end direction from a proximal end of the first tubular section **100** and can be inserted between the lock adapter **28** and the nozzle section **26**, the tubular connecting section **54** may have, on an outer peripheral section of the tubular connecting section **54**, a male screw section **64** capable of being screwed with the female screw section **32** of the lock adapter **28**, in a state where the syringe cap **10** is mounted to the syringe body **14** by screwing between the male screw section **64** of the tubular connecting section **54** and the female screw section **32** of the lock adapter **28**, and the cap body **34** is located at the first position, the mounting tube section **38** may be inserted between the tubular connecting section **54** and the nozzle section **26**, and the sealing section **42** may liquid-tightly seal the drug discharge port **26***a*, and when the male screw section **64** of the tubular connecting section **54** is screwed with the female screw section **32** of the lock adapter **28** in a state where the syringe cap **10** is removed from the syringe body **14**, the cap body **34** may be displaced from the first position to the second position by the abutting section **46** of the cap body **34** being pushed by the distal end section of the nozzle section **26**.
(89) The above embodiment discloses a syringe assembly **22** including the syringe cap **10** described above and a syringe outer tube **16** constituting the syringe body **14** and being capable of accommodating a drug M.
(90) The above embodiment discloses a prefilled syringe **12** including the syringe assembly **22** described above, a drug M filled in the syringe outer tube **16**, and a gasket **18** liquid-tightly slidable

in the syringe outer tube **16** in an axial direction.

(91) The detailed description above describes embodiments of a syringe cap, a syringe assembly, and a prefilled syringe, in which the syringe cap is detachably attachable to a syringe body that includes a body section capable of internally accommodating a drug, and a nozzle section protruding in a distal end direction from a distal end section of the body section and having a drug discharge port at a distal end. The invention is not limited, however, to the precise embodiments and variations described. Various changes, modifications and equivalents may occur to one skilled in the art without departing from the spirit and scope of the invention as defined in the accompanying claims. It is expressly intended that all such changes, modifications and equivalents which fall within the scope of the claims are embraced by the claims.

Claims

1. A syringe cap configured to be detachable from and attachable to a syringe body, the syringe body including a body section configured to accommodate a drug, and a nozzle section protruding in a distal end direction from a distal end section of the body section and having a drug discharge port at a distal end, the syringe cap comprising: a cap body; a cover member having a tubular shape and covering the cap body; the cap body includes: a mounting section including a sealing section that liquid-tightly seals the drug discharge port and being mountable to the nozzle section; and a viewing section located on a distal end side from the mounting section; the cover member includes: an engagement section provided on an inner peripheral surface of the cover member, and configured to engage with the cap body to prevent detachment of the cap body in a proximal end direction from the cover member; a first tubular section being substantially opaque, and having an opening at a distal end and a first space capable of internally housing the viewing section; and a second tubular section extending in a distal end direction from the opening of the first tubular section; the second tubular section includes: an annular peripheral wall section extending in a distal end direction from the opening of the first tubular section and having an outer diameter smaller than an outer diameter of the first tubular section; a distal end wall provided at a distal end of the annular peripheral wall section; a second space defined by the annular peripheral wall section, the distal end wall, and the opening of the first tubular section; and a tapered section protruding from an outer peripheral surface of the annular peripheral wall section toward a vicinity of an outer edge of the opening of the first tubular section so as to be inclined outward in a proximal end direction; at least the tapered section and the annular peripheral wall section of the second tubular section are substantially transparent; the cap body is configured to be displaced, along a center axis of the cover member in the cover member, from a first position where the viewing section is located in the first space of the first tubular section to a second position where the viewing section protrudes in a distal end direction from the distal end of the first tubular section to be arranged in the second space; an outer peripheral section of the viewing section is not visible when the cap body is at the first position, and the outer peripheral section of the viewing section becomes visible when the cap body is at the second position; the sealing section is capable of liquid-tightly sealing the nozzle section in a state where the cap body is located at the first position; in a state where the cap body is arranged at the first position, when the mounting section of the syringe cap removed from the syringe body is brought close to the nozzle section of the syringe body, the mounting section is pushed in a distal end direction by the nozzle section of the syringe body so as to displace the cap body from the first position to the second position; and the tapered section has, on an outer peripheral surface of the tapered section, a plurality of stepped sections continuously arranged along the center axis of the cover member, so as to gradually increase a protruding amount of the tapered section from an outer peripheral surface of the annular peripheral wall section toward a vicinity of the outer edge of the opening of the first tubular section.

2. The syringe cap according to claim 1, wherein each of the plurality of stepped sections has a first

surface, a bent section located at an outer edge of the first surface, and a second surface extending in a proximal end direction from the bent section; a first angle formed by the first surface and a parallel straight line parallel to the center axis of the cover member is larger than a taper angle formed by the parallel straight line and a virtual straight line passing through a distal end and a proximal end of the tapered section; and a second angle formed by the second surface and a parallel straight line parallel to the center axis of the cover member is smaller than the taper angle.

3. The syringe cap according to claim 1, wherein each of the plurality of stepped sections has a first surface, a bent section located at an outer edge of the first surface, and a second surface extending in a proximal end direction from the bent section; an angle formed by the first surface and a parallel straight line parallel to the center axis of the cover member is larger than a taper angle formed by the parallel straight line and a virtual straight line passing through a distal end and a proximal end of the tapered section; and the second surface extends parallel to the center axis.

4. The syringe cap according to claim 2, wherein the taper angle is 35° to 55°.

5. The syringe cap according to claim 4, wherein the first angle is 60° to 90°, and the second angle is 30° or less.

6. The syringe cap according to claim 2, wherein each of the plurality of stepped sections includes an inclined surface extending while being inclined outward from a proximal end of the second surface and continuing to the inner edge of the first surface of a stepped section adjacent on a proximal end side.

7. The syringe cap according to claim 2, wherein the bent section has a curved surface protruding outward.

8. The syringe cap according to claim 2, wherein the second tubular section is substantially transparent as a whole and has a distal end curved surface protruding outward between an outer edge of the distal end wall and the annular peripheral wall section, and a curvature radius of the distal end curved surface is 0.3 mm to 1.0 mm.

9. The syringe cap according to claim 1, wherein the cover member includes a tubular section constituting at least an outer peripheral section of the first tubular section and being substantially opaque, and a distal end cover member having the second tubular section and being substantially transparent.

10. The syringe cap according to claim 9, wherein the distal end cover member has an extending inner tube section that extends from a proximal end of the annular peripheral wall section into the tubular section and forms an inner peripheral section of the first tubular section; and an inner diameter of the distal end cover member is substantially uniform from a distal end of the annular peripheral wall section to a proximal end of the extending inner tube section.

11. The syringe cap according to claim 1, wherein the syringe body includes a lock adapter having a tubular shape, having a female screw section on an inner peripheral surface, and covering an outer peripheral section of the nozzle section; the mounting section is a mounting tube section having a tubular shape and being capable of accommodating the nozzle section; the mounting tube section includes the sealing section located at a distal end section of the mounting tube section and an abutting section located at a proximal end section of the mounting tube section and being capable of abutting with a distal end section of the nozzle section; the cover member has a tubular connecting section that extends in a proximal end direction from a proximal end of the first tubular section and is capable of being inserted between the lock adapter and the nozzle section; the tubular connecting section has a male screw section capable of being screwed with the female screw section of the lock adapter, on an outer peripheral section of the tubular connecting section; in a state where the syringe cap is mounted to the syringe body by screwing between the male screw section of the tubular connecting section and the female screw section of the lock adapter, and the cap body is located at the first position, the mounting tube section is inserted between the tubular connecting section and the nozzle section, and the sealing section liquid-tightly seals the drug discharge port; and when the male screw section of the tubular connecting section is screwed with

the female screw section of the lock adapter in a state where the syringe cap is removed from the syringe body, the cap body is displaced from the first position to the second position by the abutting section of the cap body being pushed by the distal end section of the nozzle section.

12. A syringe assembly comprising: the syringe cap according to claim 1; and a syringe outer tube constituting the syringe body and configured to accommodate the drug.

13. A prefilled syringe comprising: the syringe assembly according to claim 12; the drug filled in the syringe outer tube; and a gasket liquid-tightly slidable in the syringe outer tube in an axial direction.

14. A syringe cap configured to be detachable from and attachable to a syringe body, the syringe cap comprising: a cap body, the cap body includes: a mounting section including a sealing section configured to seal a drug discharge port on the syringe body and configured to be mounted to a nozzle section of the syringe body; and a viewing section located on a distal end side from the mounting section; a cover member having a tubular shape and covering the cap body, the cover member includes: an engagement section provided on an inner peripheral surface of the cover member, and the cover member being configured to engage with the cap body; a first tubular section having an opening at a distal end and a first space capable of internally housing the viewing section; and a second tubular section extending in a distal end direction from the opening of the first tubular section; the second tubular section includes: an annular peripheral wall section extending in a distal end direction from the opening of the first tubular section and having an outer diameter smaller than an outer diameter of the first tubular section; a distal end wall provided at a distal end of the annular peripheral wall section; a second space defined by the annular peripheral wall section, the distal end wall, and the opening of the first tubular section; and a tapered section protruding from an outer peripheral surface of the annular peripheral wall section toward a vicinity of an outer edge of the opening of the first tubular section so as to be inclined outward in a proximal end direction; the tapered section has, on an outer peripheral surface of the tapered section, a plurality of stepped sections continuously arranged along the center axis of the cover member, so as to gradually increase a protruding amount of the tapered section from an outer peripheral surface of the annular peripheral wall section toward a vicinity of the outer edge of the opening of the first tubular section; and wherein the cap body is configured to be displaced, along a center axis of the cover member in the cover member, from a first position where the viewing section is located in the first space of the first tubular section to a second position where the viewing section protrudes in a distal end direction from the distal end of the first tubular section to be arranged in the second space.

15. The syringe cap according to claim 14, wherein at least the tapered section and the annular peripheral wall section of the second tubular section are transparent.

16. The syringe cap according to claim 14, wherein an outer peripheral section of the viewing section is not visible when the cap body is at the first position, and the outer peripheral section of the viewing section becomes visible when the cap body is at the second position; the sealing section configured to be seal the nozzle section of the syringe body in a state where the cap body is located at the first position; in a state where the cap body is arranged at the first position, when the mounting section of the syringe cap removed from the syringe body is brought close to the nozzle section of the syringe body, the mounting section is pushed in a distal end direction by the nozzle section of the syringe body so as to displace the cap body from the first position to the second position.

17. The syringe cap according to claim 16, wherein each of the plurality of stepped sections has a first surface, a bent section located at an outer edge of the first surface, and a second surface extending in a proximal end direction from the bent section; a first angle formed by the first surface and a parallel straight line parallel to the center axis of the cover member is larger than a taper angle formed by the parallel straight line and a virtual straight line passing through a distal end and a proximal end of the tapered section; and a second angle formed by the second surface and a

parallel straight line parallel to the center axis of the cover member is smaller than the taper angle.
18. A syringe assembly, the syringe assembly comprising: a syringe body, the syringe body including a body section configured to accommodate a drug, and a nozzle section protruding in a distal end direction from a distal end section of the body section and having a drug discharge port at a distal end; a syringe cap configured to be detachable from and attachable to the syringe body, the syringe cap comprising: a cap body, the cap body includes: a mounting section including a sealing section configured to seal a drug discharge port on the syringe body and configured to be mounted to a nozzle section of the syringe body; and a viewing section located on a distal end side from the mounting section; a cover member having a tubular shape and covering the cap body, the cover member includes: an engagement section provided on an inner peripheral surface of the cover member, and the cover member being configured to engage with the cap body; a first tubular section having an opening at a distal end and a first space capable of internally housing the viewing section; and a second tubular section extending in a distal end direction from the opening of the first tubular section; the second tubular section includes: an annular peripheral wall section extending in a distal end direction from the opening of the first tubular section and having an outer diameter smaller than an outer diameter of the first tubular section; a distal end wall provided at a distal end of the annular peripheral wall section; a second space defined by the annular peripheral wall section, the distal end wall, and the opening of the first tubular section; and a tapered section protruding from an outer peripheral surface of the annular peripheral wall section toward a vicinity of an outer edge of the opening of the first tubular section so as to be inclined outward in a proximal end direction; and the tapered section has, on an outer peripheral surface of the tapered section, a plurality of stepped sections continuously arranged along the center axis of the cover member, so as to gradually increase a protruding amount of the tapered section from an outer peripheral surface of the annular peripheral wall section toward a vicinity of the outer edge of the opening of the first tubular section; and wherein the cap body is configured to be displaced, along a center axis of the cover member in the cover member, from a first position where the viewing section is located in the first space of the first tubular section to a second position where the viewing section protrudes in a distal end direction from the distal end of the first tubular section to be arranged in the second space.
19. A prefilled syringe comprising: the syringe assembly according to claim 18; the drug filled in the syringe body; and a gasket liquid-tightly slidable in the syringe body in an axial direction.